TD/TP 2
Fundamental Mathematical Concepts

2.4 Let $\{e_t\}$ be a zero mean white noise process. Suppose that the observed process is $Y_t = e_t + \theta e_{t-1}$, where θ is either 3 or 1/3.

a) Find the autocorrelation function for $\{Y_t\}$ both when $\theta = 3$ and when $\theta = 1/3$.

b) You should have discovered that the time series is stationary regardless of the value of θ and that the autocorrelation functions are the same for $\theta = 3$ and $\theta = 1/3$. For simplicity, suppose that the process mean is known to be zero and the variance of Y_t is known to be 1. You observe the series $\{Y_t\}$ for $t = 1, 2, ..., n$ and suppose that you could determine which value of θ is correct (3 or 1/3) based on the estimate of ρ_k? Why and why not?

2.5 Suppose $Y_t = 5 + 2t + X_t$, where $\{X_t\}$ is a zero-mean stationary series with autocovariance function γ_k.

a) Find the mean function for $\{Y_t\}$.

b) Find the autocovariance function for $\{Y_t\}$.

c) Is $\{Y_t\}$ stationary? Why or why not?

2.8 Suppose that $\{Y_t\}$ is stationary with autocovariance function γ_k. Show that for any fixed positive integer n and any constant $c_1, c_2, ..., c_n$, the process $\{W_t\}$ defined by $W_t = c_1 Y_t + c_2 Y_{t-1} + ... + c_n Y_{t-n+1}$ is stationary. (Note that Exercise 2.7 is a special case of this result.)

2.9 Suppose $Y_t = \beta_0 + \beta_1 t + X_t$, where $\{X_t\}$ is a zero-mean stationary series with autocovariance function γ_k and β_0 and β_1 are constants.

a) Show that $\{Y_t\}$ is not stationary but $W_t = \nabla Y_t = Y_t - Y_{t-1}$ is stationary.

b) In general, show that if $Y_t = \mu_t + X_t$, where $\{X_t\}$ is a zero-mean stationary series and μ_t is a polynomial in t of degree d, then $\nabla^m Y_t = \nabla(\nabla^{m-1} Y_t)$ is stationary for $m \geq d$ and nonstationary for $0 \leq m < d$.

2.12 Suppose that $Y_t = e_t - e_{t-12}$. Show that $\{Y_t\}$ is stationary and that, for $k > 0$, its autocorrelation function is nonzero only for lag $k = 12$.

2.14 Evaluate the mean and the covariance function for each of the following processes. In each case, determine whether or not the process is stationary.

a) $Y_t = \theta_0 + te_t$

b) $W_t = \nabla Y_t$, where Y_t is as given in part (a).

c) $Y_t = e_t e_{t-1}$. (You may assume that $\{e_t\}$ is normal white noise.)

2.17 Let $\{Y_t\}$ be stationary with autocovariance function γ_k, Let $\bar{Y} = \frac{1}{n}\sum_{t=1}^{n} Y_t$. Show that

$$Var(\bar{Y}) = \frac{\gamma_0}{n} + \frac{2}{n}\sum_{k=1}^{n-1}\left(1 - \frac{k}{n}\right)\gamma_k$$

$$= \frac{1}{n}\sum_{k=-n+1}^{n-1}\left(1 - \frac{|k|}{n}\right)\gamma_k.$$

2.23 Two process $\{Z_t\}$ and $\{Y_t\}$ are said to be **independent** if for any time points $t_1, t_2, ..., t_m$ and $s_1, s_2, ..., s_n$ the random variable $\{Y_{s_1}, Y_{s_2}, ..., Y_{s_n}\}$. Show that if $\{Z_t\}$ and $\{Y_t\}$ are independent stationary processes, then $W_t = Z_t + Y_t$ is stationary.

2.26 Define the function $\Gamma_{t,s} = \frac{1}{2}E[(Y_t - Y_s)^2]$. In geostatistics, $\Gamma_{t,s}$ os called the *semi-variogram*.

a) Show that for a stationary process $\Gamma_{t,s} = \gamma_0 - \gamma_{|t-s|}$.

b) A process is said to be *intrinsically stationary* if $\Gamma_{t,s}$ depends only on the time difference $|t-s|$. Show that the random walk process is intrinsically stationary.

2.27 For a fixed, positive integer r and constant ϕ, consider the time series defined by $Y_t = e_t + \phi e_{t-1} + \phi^2 e_{t-2} + ... + \phi^r e_{t-r}$.

a) Show that this process is stationary for any value of ϕ.

b) Find the autocorrelation function.